

U.S. Food and Drug Administration Regulation of Prosthetic Research, Development, and Testing

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ABSTRACT

The purpose of this article is to present an overview of U.S. Food and Drug Administration (FDA) regulatory requirements to inform those working to develop and test new prosthetic devices and controls. This article summarizes the risk classification for medical devices, which is based on level of risk control necessary to provide reasonable assurance of safety and effectiveness. Most, but not all, prosthetic components and controls are classified as class I (minimal risk and general controls for risks). Nearly, all class I devices are exempt from premarket notification by FDA. Although data on safety and clinical effectiveness may not be required to obtain FDA clearance to market in the United States, most class I devices, manufacturers, or researchers may still wish to conduct studies using these devices. Studies of low risk to study subjects are called “nonsignificant risk investigational devices studies” and are regulated by Federal codes but do not ordinarily require approval of an investigational device exemption from FDA. Class II devices represent higher risk requiring special controls in addition to the general controls for class I devices. Most, but not all, class II devices require premarket notification (510k) submission, and these typically compare intended use and technology of the device to those of a device already marketed through the 510(k) process. Limited clinical data are occasionally needed for some class II devices. Class III devices are those with the greatest risk, and a premarket approval application, generally with prospective clinical data to demonstrate safety and effectiveness, is needed. This article presents highlights of these regulatory requirements. (*J Prosthet Orthot.* 2010;22:121–126.)

KEY INDEXING TERMS: prosthetics, United States Food and Drug Administration, medical devices, government regulation

Prostheses are considered “medical devices,” and as such are subject to regulation by the U.S. Food and Drug Administration (FDA), Center for Devices and Radiological Health (CDRH). Although there have been articles published in the literature describing FDA regulation of medical devices, these articles tend to focus on devices of significant risk, including implants, and do not provide guidance specific to the prosthetic industry.^{1–4} The FDA/CDRH supervises the approval of all new medical devices, defines their category of risk, reviews research studies involving significant risks to human subjects, and helps determine the kind of evidence or investigational studies, if any, is needed to

provide reasonable assurance of safety and effectiveness for approval to market the device in the United States. All prosthetic devices must comply with Federal regulations through FDA/CDRH processes before they can be marketed within the United States as required by 21 CFR Parts 800-1299. Before approval for marketing by the FDA/CDRH, a medical device under investigation with human trials is called an “investigational device.”

The FDA regulates investigational devices through Federal Regulations (21 CFR Part 812) and by the Federal Food, Drug and Cosmetics Act. Therefore, prosthetic manufacturers and designers need to understand FDA regulations and requirements before performing clinical studies with prosthetic devices. Thus, the purpose of this article is to present an overview of the FDA's regulatory requirements to inform those working to develop and test new prosthetic devices and controls. This article focuses on requirements for prosthetic devices that mostly fall into class I or class II and often involve studies without significant risks to the study subjects.

PATHS TO MARKETING MEDICAL DEVICES

Before any new medical device can be marketed in the United States, manufacturers must either submit FDA premarket review in one of two forms, i.e., premarket approval (PMA) application or premarket notification [510(k)] submission, or must document that their device meets the criteria for exemption from such submissions. Most existing devices have already been classified by FDA along with the type of submission, if any, that is required. For example, physical

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medicine devices are listed along with other pertinent information in 21 CFR Part 890. The primary office within CDRH responsible for premarket review is the Office of Device Evaluation.⁵ Because there are certain exceptions in regulations and devices may contain a complex set of components and technology and there may be multiple clinical indications that vary with population, CDRH suggests that investigators discuss concerns with the Office of Device Evaluation early in the process.

DEVICE RISK CLASSIFICATION

According to the FDA/CDRH, new devices are categorized as low risk (class I), moderate risk (class II), or high risk (class III) on the basis of the controls needed to assure their safety and effectiveness at the level of risk for the intended use and indications for use for the device.⁵ Figure 1 shows the classification of selected physical medicine devices.

Manufacturers can obtain a classification ruling for a new device (not already listed in 21 CFR Part 890) by submitting a request for device classification under section FDA 513(g) of the Food, Drug and Cosmetics Act. According to 21 CFR Part 812 (Parts 890.3420 and 890.3025), the FDAs product Classification listings, external limb prosthetic components, and accessories are defined as physical medicine devices and are generally considered class I devices.⁶ Class I devices are considered low risk. They have minimal potential for harm and are specifically defined by the FDA as not intended to be for use in supporting or sustaining life, of importance in preventing impairment to human life, and may not present a potential unreasonable risk of illness or injury.⁷ For a new device, the section 513(g) request provides a formal mechanism for the FDA to determine a device class.

Device classes are further defined by the level of controls necessary to reasonably assure safety and effectiveness. Class I devices require what the FDA codes call “general controls” to meet this standard. General controls are provisions of the May 28, 1976, Medical Device Amendments to the Food, Drug and Cosmetic Act that provide the FDA with the means of regulating devices to ensure their safety and effectiveness. General controls apply to all medical devices, regardless of level of risk; they relate to adulteration; misbranding; device registration and listing; premarket notification; banned devices; notification, including repair, replacement, or refund; records and reports; restricted devices; and Good Manufac-

turing Practices.⁷ Some class I devices are exempt from Good Manufacturing Practices including canes and mechanical walkers.⁶ Class II devices are those for which general controls alone are insufficient to assure safety and effectiveness, and existing methods, termed special controls, are available to provide such assurances. Special controls may include special labeling requirements, mandatory performance standards, and postmarket surveillance.⁸ Functionally, class II devices are often referred to as moderate or intermediate risk. Examples of class II devices include infusion pumps, endoscopes, electric wheelchairs, and surgical drapes.

Class III devices are those considered highest risk. Class III devices are usually those that support or sustain human life, are of substantial importance in preventing impairment of human health, or which present a potential, unreasonable risk of illness or injury. From the perspective of controls, class III devices are those for which insufficient information exists to assure safety and effectiveness solely through general or special controls. Therefore, a PMA process of scientific review is usually required to prove safety and effectiveness of class III devices.⁸

FDA MARKETING REQUIREMENTS VARY BY DEVICE CLASSIFICATION

The route to marketing a new medical device varies based on the FDA/CDRH Classification of the device. Table 1 provides a synopsis of the FDA approval process and requirements by class of device.⁹

CLASS I DEVICES

The vast majority of class I devices are considered exempt from premarket review, neither a PMA application nor a Premarket Notification [510(k)] submission, and manufacturers need to be submitted for FDA approval. The collection of clinical data is not required by FDA for these devices. However, as stated previously, class I devices are subject to general controls requirements, such as medical device listing, labeling standards, and registration and device listing. They must also keep records and report when the device causes or contributes to death or serious injury.

If a class I prosthetic device operates using a different fundamental scientific technology than a legally marketed device in that generic type of device or if it is being marketed for a new use, it does require a premarket notification submission [510(k)] (see below). Thus, a prosthetic device that introduces a new technology may require 510(k) submission.

CLASS II DEVICES

Most class II devices, and class I devices that are not exempt, require a premarket notification [510(k)] submission. The purpose of a 510(k) submission is to demonstrate that the new device, though it may be technologically different, is equally safe and effective (substantially equivalent) as another, similar class II device already legally marketed in the United States. The existing legally marketed device to which

Class I	Class II	Class III
• External limb components • Socket components • Foot drop orthosis • Other limb braces	• Most hearing aids • C-Leg microprocessor knee • Powered external limb overload warning device • Powered wheelchair	• Stair climbing wheelchair

Figure 1. Classification of physical medicine devices.

Table 1. Synopsis of FDA approval of new medical device

Device class	FDA approval process	Clinical data required
Class I	Usually exempt by FDA specification from all premarket submissions and/or good manufacturing practices regulation Must meet FDA general controls requirements, labeling standards, medical device listing, and establishment registration on file with the FDA	None
Class II	Unless exempt, Premarket Notification (510k) submission must demonstrate that the device is as safe and effective, i.e., substantially equivalent (SE), as an existing legally marketed device that is not subject to PMA	Not usually required (as below)
Class III	Premarket approval (PMA) application must demonstrate rigorous scientific and regulatory review to evaluate safety and effectiveness	Controlled studies Partially controlled studies Uncontrolled studies Well-documented case histories Reports of significant human experience

the proposed new one is compared is called a “predicate” device.⁴ For example, many medical devices are predicated on previous product design in which the materials are well understood, and the interactions of those materials with the human body are known.¹⁰ When the FDA determines that the device is substantially equivalent, it is cleared for marketing.⁵ A few specified class II devices are exempt from the requirement for 510(k) submission under the FDA Modernization Act of 1997. Examples of these class II devices exempt from 510(k) premarket notification include external assembled lower limb prostheses (21 CFR Part 890.3500), audiometers, water-circulating hot or cold packs, and clinical mercury thermometers.¹¹

When a device has a new intended use, including for a new patient population, or it uses new technological features, the FDA may require the submission of clinical data to support the determination substantial equivalence (SE). However, a large percentage of class II devices are found to be substantially equivalent on the basis of comparative bench performance and preclinical data alone, and only 10% to 15% of 510(k) submissions contain or require clinical data to demonstrate SE.⁵ If on review, the device is not found to be substantially equivalent, it is typically categorized as a class III device (<http://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/HowtoMarketYourDevice/PremarketSubmissions/PremarketNotification510k/default.htm>). The manufacturer has the option to petition the FDA to obtain a de novo class II device classification if they believe that their device is both novel and of low risk that been be mitigated through special controls.⁶ In their application, submitters must identify the risks and benefits of the device, itemize the applicable controls, and recommend a device classification as either class I or class II. The FDA will review the request and make the final device classification.

CLASS III DEVICES

Because of their higher risk, class III devices must undergo greater premarket scrutiny than class II devices. Device

manufacturers must submit a PMA application for all class III devices and must provide reasonable assurance of the device's safety and effectiveness. Reasonable assurance means the provision of valid scientific clinical evidence derived from clinical studies of the device, its effectiveness, and safety. The percentage of medical devices classified as high risk constitutes less than 5% of the devices regulated by FDA/CDRH.⁵

A few class III devices are exempt from PMA based on SE to existing devices on the market before the passage of the medical device amendments in 1976. One example is the external defibrillator, and a 510(k) is the route to market.¹² In April 2009, the FDA issued a Federal Register notice requiring manufacturers of the 23 devices in this category to provide evidence to support either reclassification to class II or a call for PMAs, if the devices remain in class III (Docket No. FDA-2009-M-0101).

A regulatory path similar to the PMA, which can apply to class II or class III devices, is the humanitarian device exemption (HDE) that applies to humanitarian use devices. The HDE application is similar to the PMA application, but the approval is based on whether the probable benefit outweighs the risk of injury from its use. This is different from a PMA approval in which a reasonable assurance of safety and effectiveness is required. A Humanitarian Use Device is defined in 21 CFR 814.3(n) as a “medical device intended to benefit patients in the treatment or diagnosis of a disease or condition that affects or is manifested in fewer than 4,000 individuals in the United States per year.” If a comparable device is already on the market [by an approved PMA or 510(k)], then FDA cannot grant a HDE. A number of prosthetic devices may qualify for a HDE application.

REGULATIONS OF RESEARCH WITH MEDICAL DEVICES

Regulations concerning the need for FDA approval of an investigational device exemption (IDE) for clinical studies with medical devices differ depending on whether there is

significant risk or nonsignificant risk for human subjects. All medical device studies with human subjects require an institutional review board (IRB) approval, study-subject protection, and informed consent of subjects. It is important to note that the clinical study of a device is considered as either a significant risk or a nonsignificant risk study. Sponsors (including investigators at academic institution) of a proposed device study can seek determination of risk status from their IRB. This is in keeping with the Federal codes that require FDA to pursue the “least burdensome means” to establish device safety and efficacy.² However, the FDA can also determine whether a device study is significant or nonsignificant risk. FDA published a guidance document for IRBs on significant risk and nonsignificant risk device studies, and this guidance includes lists of devices that are considered as significant risk.¹³

The IRB is an administrative committee or board whose role it is to review all research involving human subjects. The IRB has authority to approve, disapprove, or require modifications to a research protocol. According to the U.S. Department of Health and Human Services Office for Research Protections IRB Guidebook, an institution conducting research on human subjects must either establish an IRB or designate one from another institution.¹⁴ Device manufacturers and developers often use what is called a central or independent IRB, which are often run as for-profit business.

An IRB can serve as the FDAs surrogate for review, approval, and continuing review of non-significant risk (NSR) device studies. These types of studies may in fact be conducted without the FDAs knowledge. Unless the FDA informs the sponsors otherwise or the device is banned, NSR studies are considered to have approved IDE applications. However, in all cases of NSR device, studies must have IRB approval, the sponsor must comply with “abbreviated IDE requirements” (as required by 21 CFR 812.2), and the informed consent and IRB regulations (21 CFR Parts 50 and 56). The FDAs abbreviated requirements for nonsignificant risk investigational studies have been applied to research regarding upper limb prosthetics. A summary of these requirements are provided in Table 2.

An IDE allows unapproved class III devices to be used in clinical studies to collect the safety and effectiveness data required to support either a PMA application or a premarket notification [510(k)] submission to FDA.¹¹ It is also needed when devices are used in research studies that may not be directed toward marketing. In addition, class I or class II devices can be used in significant risk studies and require an approved IDE. Although data on safety and clinical effectiveness may not be required for new class I and II devices to obtain FDA approval, manufacturers may still wish to conduct research using these devices for a number of reasons. First, the FDA may in some instances require data demonstrating safety and effectiveness. Second, the manufacturer may wish to conduct studies to evaluate device performance, refine the device design, or compare to existing technologies.

Table 2. Abbreviated requirements for sponsors of nonsignificant risk device clinical studies: 21 CFR812.2 (b)

Device labeling	Label device with name and place of business of manufacturer, packer, or distributor, quantity of contents, contraindications, hazards, adverse effects, interfering substances or devices, and statement of caution:
IRB approval	Ensure investigators obtain and maintain IRB approval
Informed consent	Assure informed consent is obtained from subjects
Investigator records and reports	Ensure investigators maintain records and reports (see Table 3)
Monitoring requirements	Comply with monitoring requirements, such as those in investigational plan, conditions imposed by IRB, and unanticipated device effects reporting
Sponsor records and reports	Maintain required records and reports, such as name and intended use of the device, objectives of the investigation; explanation of non-significant risk device, name and address of each investigator and IRB, records regarding complaints, and adverse device effects
Prohibition on promotion	Comply with prohibition on promoting, test marketing, commercializing, or representing as safe or effective a device, before FDA approval promotion

Studies of low-risk devices used in a low-risk study are called “nonsignificant risk investigational devices studies” or NSR studies. Although NSR studies are subject to regulation by the FDA, they are considered to have approved applications for IDEs and have abbreviated requirements for reporting. An overview of regulatory requirements for research is shown in Table 3.

The differences in reporting requirements between significant risk and nonsignificant risk studies apply both to the sponsors (who are typically the device manufacturers) and the investigators involved in the study.¹⁵ Table 4 compares the differences in reporting and record-keeping responsibilities for investigators.

The FDA encourages early interaction for those considering initiating an investigation that may require an IDE. The interaction can take the form of a pre-IDE submission and a pre-IDE meeting. The pre-IDE submission may contain draft clinical protocols and proposals for preclinical testing. The FDA will provide preliminary and informal review and comment to facilitate the IDE application.

IMPLICATIONS FOR PROSTHETIC DEVELOPMENT, PRACTICE, AND REIMBURSEMENT

Health insurers and clinicians need scientific evidence to guide practice and reimbursement policies. Because prosthetic devices are typically considered class I devices, most of the new devices that have come on the market in the past few decades have been exempt from FDA regulatory requirements

Table 3. Overview of regulatory requirements for research with new medical devices

Device class	FDA requirement
Class I	IRB makes determination of nonsignificant risk study Study subject to abbreviated reporting requirements
Class II	IRB makes determination of nonsignificant risk or FDA grants investigational device exemption (IDE)
Class III	FDA grants investigational device exemption (IDE)

of PMA and premarket notification [510(k)]. Therefore, the majority of new prosthetics have entered the commercial marketplace fulfilling requirements for general controls. The paucity of research on prosthetic devices may be explained, in part, by the fact that marketing studies demonstrating safety and effectiveness are not required by Federal Law for most class I devices whose safety and effectiveness can be reasonably assured by general controls.

Investigational medical devices are not generally reimbursed by third parties. However, the device manufacturer is permitted to recover costs of the experimental device and the medical care used during the investigation.

Today's advances in prosthetic engineering and controls are introducing a host of new technologies and uses. Many of these new developments will require premarket notification [510(k)] submissions, and the FDA may need the submission of clinical data to determine safety and effectiveness. Prosthetic manufacturers and developers should be familiar with the regulations governing nonsignificant risk device studies, so that they can perform studies evaluating the risks and benefits of new devices and technology. Ultimately, the conduct of such studies will benefit manufacturers and consumers.

Table 4. Investigator responsibilities for significant risk and non-significant risk device studies

Significant risk study	Nonsignificant risk study
Informed consent An investigator is responsible for obtaining informed consent under 21 CFR Part 50	Same as significant risk study
Supervision of device use Can permit use of the device only with subjects under his/her supervision cannot not supply the device to any person not authorized under the IDE	None specified
Required records Each subject's case history Each subject's exposure to the device Correspondence including required reports Receipt, use, or disposition of the investigational device Protocol Protocol deviations Any other records required by FDA for the particular investigation	Each subject's case history Each subject's exposure to the device
Required reports Unanticipated adverse device effects Withdrawal of IRB approval Informed consent Annual progress reports Final reports Deviations from the investigational plan Final report Other reports as requested by IRB or FDA	Unanticipated adverse device effects Withdrawal of IRB approval Informed consent Other Reports as requested by IRB or FDA
Financial disclosure Disclose to the sponsor sufficient information to allow IDE applicant (or sponsor) to submit certification or disclosure of financial interests under 21 CFR 54 Update information if relevant changes occur	If data are submitted in a marketing application, then same as significant risk study
Device disposal Return to sponsor any remaining supply of the device or dispose of device as sponsor directs	None specified

FDA encourages those planning a study using an IDE to meet with CDRH staff members before the IDE is submitted for review. An explanation of the pre-IDE program can be found at <http://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/GuidanceDocuments/ucm126600.htm>.

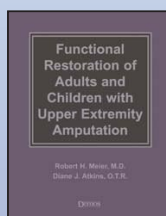
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